

ShuttleLab — Full Documentation

Measure. Improve. Dominate.

Live site: <https://shuttle-lab-mu.vercel.app> Last updated: 13 June 2026

1. What ShuttleLab is

ShuttleLab is the private training platform for our badminton team — one coach (Ahmed Salah) and his players. It runs in any browser on a phone or laptop and can be installed like a real app. It covers the full coaching loop:

- **Measure** — log fitness-test results and watch progress over a season.
- **Plan** — season targets, dated training phases, and a tournament calendar per player.
- **Train** — a professional 3D drill builder (the **Pro Court**) for singles, doubles and mixed, plus a drill library.
- **Analyse** — a film room where the coach pins notes, drawings, slow-motion and his own voice onto match videos.
- **Support** — the Mind Room, a calming space that looks after the players' heads, not just their legs.

Everyone signs in with their own account. The coach sees the whole team; players see themselves (and what the coach shares with them).

Install it like an app

Open the site on a phone → browser menu → **Add to Home Screen**. It gets the ShuttleLab icon, opens full-screen, and the app shell works even with a weak connection. Stay signed in across visits — re-login is only needed on a new device or after signing out.

2. Accounts & roles

Role	Who	Tabs
Coach	Ahmed Salah	Team · Training · Library · Matches · Leaderboard · Mind Room
Player	H, KOKA, Mahmoud, Alya El Ghandour	Dashboard · Log · Training · Library · Videos · Leaderboard · Mind Room

- Sign-in uses a **private email + password** per person (the emails are internal IDs like `name@shuttlelab.app`, not real mailboxes).
 - The team roster (names, roles, photos) lives in the protected database — it is **not** visible in the site's code, and no emails ship to the browser.
 - Players add their **own** test results (the coach no longer has a Log tab); the coach sees everything everywhere else.
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3. Page-by-page guide

3.1 Dashboard (players; coach sees it per player via Team)

The player's progress home:

- **Trend chart** per test with a dashed gold **personal-best line** and a gold dot on the PB session.
- **Personal bests** — all-time best per test, the date, improvement since the first session, and a green **NEW PB** badge when the latest session breaks one ("N new PBs this session" in the header).
- **Goals** — the coach sets a target per test; progress shows against it.
- **Compare** — current values vs the previous session.
- **Export PDF** (*coach only*) — a printable report: personal bests, trend charts, results table, goals and signature lines. Use the browser's "Save as PDF" (allow pop-ups).

3.2 Team (coach)

The whole squad at a glance — every player's latest results per test, with tap-through to each player's full Dashboard.

3.3 Log (players)

Record a test session: pick the date, enter results. Multi-attempt tests give three boxes — the app keeps the **best and the average** automatically.

The six tests

Test	Unit	Better is	Attempts
Beep test	level	higher	single
Vertical jump	cm	higher	3
Standing long jump	cm	higher	3
In-out agility	s	lower	3
Forward & backward court	s	lower	3
Forehand & backhand defense	s	lower	3

(Tests are configurable in [js/config.js](#).)

3.4 Training — the season-planning hub

The coach picks a player and manages three sections; the player sees them read-only (live-updating):

1. 🎯 **Season target** — one pinned goal statement for the season.
2. 📅 **Plan phases** — dated training blocks (start → end, focus, details) stacked into a timeline, e.g. "1 Jan → 1 Feb · Base endurance."
3. 🏆 **Tournaments** — the coach **stars** events from the shared schedule for each player. The player sees only their starred events, sorted by date, with a days-until countdown and a **NEXT** badge on the soonest.

-  **Edit schedule** (coach): change any tournament's name, categories, dates or venue, add or remove events — updates the whole team's calendar. The 2026 federation schedule is pre-loaded.
-  **Traveling?** (player): players add tournaments they're playing abroad — these appear on their calendar with a  TRAVEL badge and are surfaced to the coach in a highlighted card.

4. **Messages** — a running coach → player notes thread.

3.5 Library — drills & the Pro Court

Two kinds of entries, both assignable to the **whole team or chosen players** ("Send to" picker). Players only see what's sent to them.

- **Text drills** — title, category (Shots/Footwork/Strength/Conditioning/ Tactics) and description.
- **Court drills** — built in the **Pro Court** (see section 4). Cards show the discipline and mode (e.g. "*Doubles · Pro 3D · Multi-shuttle*") with  **Open court** for everyone and Edit/Delete for the coach.
- **Classic courts** — the original Side Court and Solo Court remain available and all old saved drills still open.

3.6 Matches / Videos — the film room

Coach side (Matches):

- **Add video** by uploading a file (phone or laptop) or pasting a YouTube/direct link. Uploads go to our own private cloud storage with a live progress %, a 500 MB limit, and duplicate detection.
- Open a video to coach it:
 -  **Note here** pins a coaching point to the exact second.
 - Per point: written note, **auto-pause**, **slow-motion** (0.5× / 0.25×), **drawings** on the frame (arrows/circles, colour palette), and a **voice note** — tap  Record, speak the correction, and it plays in the player's ear automatically at that moment ( Hear coach to replay).
 - **Save & send** to the team or specific players.
- **Who's watched** — per video: ✓ with date, view counts, "Not yet"; cards show "*Seen by X/Y.*"
-  **Delete video** removes it for everyone and also cleans the stored files.
- Player uploads appear here flagged "  **Submitted by [name].**"

Player side (Videos): two sub-tabs —

- **From your coach** — assigned videos; playback fires the coach's notes, pauses, slow-mo, drawings and voice automatically.
- **My uploads** — players (e.g. abroad without the coach) upload their own clips for analysis; status shows "*Awaiting coach analysis*" → "*Coach added N coaching points.*" Players can delete their own uploads.

3.7 Leaderboard

Rankings per test (all-time best, or Latest toggle), medals for the top three. **Privacy:** the coach sees full names; players see the rankings and their own highlighted place, but other players appear anonymously.

3.8 Mind Room — the team's quiet corner

A page that's different every single time you enter:

- Tap the glowing **door** → fall into a randomly chosen world (starry night, fireflies, floating lanterns, falling petals, rain) with a breathing orb to follow. **Tap anywhere** to scatter light. **Shake the phone** for a new world. (There's a hidden surprise too — tap the breathing word five times.)
 - **A word for you** — a powerful encouragement in Coach Ahmed's voice, different each visit. The coach's **My words** jar adds his own messages to the rotation, signed by him.
 - **Hold my hand** — a wordless calming game: pop slow bubbles, each whispers a kind word.
 - **Leave a star** — a one-tap mood check-in that builds a private constellation **only that player can ever see** (stored in a private, owner-only space — not even the coach can read it).
 - **Talk to [coach]** — sends a private note straight to the coach; only the sender and the coach can read it. The coach reads these under **Messages** in his Mind Room.
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4. The Pro Court — full manual

The Pro Court is a true 3D-projected badminton court with posed athlete figures, used for **Singles, Doubles and Mixed doubles**, in two modes: **Drills** (rallies) and **Multi-shuttle** (coach-fed).

4.1 The court itself

Exact to the Laws of Badminton: 13.40 × 6.10 m, singles sidelines 0.46 m in, net 1.524 m centre / 1.55 m posts, short service lines 1.98 m from the net, doubles long service line 0.76 m from the baseline, 40 mm lines. Every net-crossing shot is automatically lifted to clear the tape — nothing ever flies through the net.

4.2 Cameras

Four angles, switchable any time — even mid-replay — with a smooth glide: 📺 **Broadcast** (TV angle, default) · **Corner** · **Side** (best for arcs and smash angles) · **Bird's-eye** (best for patterns and rotations).

4.3 Building a drill

1. Pick a **shot** and a **colour** (22-colour palette incl. dark tones).
2. **Tap where it's hit from** — the nearest player on that side becomes the hitter (shown in the hint).
3. **Tap where it lands**. The step joins the list; landings outside the legal court are tagged **OUT**; impossible chained shots get a ⚠️ "*probably unreachable*" warning (it never blocks you).
4. Optional **coaching note** per shot; tap any step in the list to watch just that step; ✕ deletes it.

Serves are law-enforced: choose a serve, tap the server's position, and the **legal diagonal service box lights up** — *long & narrow* in singles, *short & wide* in doubles/mixed. A target outside it is refused.

4.4 Positioning vs choreography (important!)

- **Set start** — tap a player, tap a spot: sets their **starting lineup**. Never part of the drill; nothing plays. (Using **Move** before any shots exist does the same thing automatically.)
- **Move** — once the rally has shots: tap a player, tap a destination — a **real choreographed movement** that runs *during* the rally, in parallel with the shuttle.
- **Recover** — one tap sends the last hitter back to base.

4.5 How the rally plays

- Players run to the shuttle, strike with the right body action — deep lunge at the net, overhead arch, an airborne **jump smash**, a small hop for the **slice/cut drop**, a low crouch in defense — with pose blending, shadows and rackets.
- **Contact points are real:** the shuttle launches from the racket — a net shot leaves from the outstretched hand in front of the lunge, an overhead from above the head — never from the player's chest.
- **Rally continuity:** when the next shot is taken from where the previous one arrives, the shuttle **never touches the floor** — the receiver runs in *during* the flight and takes it out of the air. Landing rings only appear when a rally truly ends.
- In **1v1 singles**, real rally habits are automatic: the hitter recovers to base *while* the opponent covers, and defenders crouch against smashes.

4.6 Doubles & Mixed rotation

With a 2v2 lineup and **Auto rotate** on (default):

- An **attacking shot** (smash, drop, kill...) flows that pair into **front-and-back** — in level doubles whoever is nearer the net takes the front; **in mixed the woman always takes the front and the man the rear** — while the opponents flatten **side-by-side** (left player stays left).
- A **lift or clear swaps the stances**. Rotations flow during the flight.
- Turn **Auto rotate off** to choreograph every movement yourself with Move.
- Mixed pairs are marked **W / M** on court.

4.7 Squads

In Drills, set each side from **1 to 4 players** (e.g. **3 vs 1**, 2 vs 1 pressure drills). With more than 1v1, automatic movement steps aside — the coach controls everything; the hitter is always the player nearest your tap.

4.8 Multi-shuttle

- **Place feeder** anywhere on the court (saved with the drill).
- Tap where each feed lands → tap the player's answer. The **feeder releases first and the player chases during the flight** — both meet at the exact contact point and the strike happens on arrival. No waiting, no run-up.
- **Feed rate** — Rapid / Steady / Relaxed rhythm (saved). At Rapid, several shuttles can be airborne at once, like a real basket session.

4.9 Watching

► **Play loops the drill again and again until you pause.** Speed control: Fast / Normal / Slow. ↺ restarts. Tapping a step in the list loops just that step — perfect for drilling one moment.

4.10 The shot library

Shot	Contact	Flight	Body
Low serve	low, front foot	skims the tape, just past the short line	serve stance
High serve	low, front foot	very high, drops vertically at the back	serve stance

Shot	Contact	Flight	Body
Flick serve	low, front foot	surprise arc to the back of the box	serve stance
Clear	overhead	high & deep to the baseline	overhead
Attacking clear	overhead	flatter, behind the shoulder	overhead
Drop shot	overhead	floats down before the short line	overhead
Slice / cut drop	high overhead	steep & fast, near-smash	small hop
Smash	high overhead	straight steep descent	overhead
Jump smash	highest point	steepest angle	full jump
Half smash	overhead	steep, placed	overhead
Drive	side-front, chest height	flat exchange	side hit
Push	side-front	into the mid-court gap	side hit
Net shot	arm outstretched at the tape	tumbles tight over	deep lunge
Net kill	reaching above the tape	straight down	lunge
Lift / lob	low, in front, under the shuttle	high to the rear	lunge
Block	low, in front	soft reply short	low defense

5. Privacy & security

- **Sign-in:** Firebase Authentication; passwords never appear in the code. Sessions persist per browser.
- **Roster privacy:** names/roles live in the database, readable only after sign-in; **no emails ship in the site's code.**
- **Database rules** (enforced server-side):
 - Test data, goals, drills, training and plans: any signed-in member.
 - Shared tournament schedule & the coach's gift jar: **coach-write-only.**
 - Videos: anyone signed-in may create (players upload their own clips); only the coach edits content or deletes any clip; a player may delete a clip **they** uploaded; players may only update watched-status otherwise.
 - **Mind Room mood stars:** owner-only — nobody else, including the coach.
 - **Mind Room messages:** readable only by the sender and the coach.
- **Video storage:** private Cloudflare R2 bucket. Files are reached only via short-lived signed links issued to signed-in users; upload permission is checked server-side; deleting a video also deletes its files (coach-gated).
- **Leaderboard:** players see rankings but other players' names are hidden.

6. Technical overview (for future maintainers)

Stack: plain HTML/CSS/JavaScript (ES modules, no framework) · Firebase Auth + Firestore (data) · Cloudflare R2 via Vercel serverless functions (video/voice files) · Chart.js (charts) · PWA service worker.

Hosting: GitHub [Mohamedhesham73/ShuttleLab](#) → Vercel auto-deploys [main](#). Firebase project: [shuttlelab-174d2](#).

Key files

Path	What it is
index.html	shell, fonts, Chart.js, service-worker registration
sw.js	PWA worker (network-first, offline fallback)
js/config.js	Firebase config, tests, tournaments, upload limit, coach contact
js/app.js	auth gate, roster load, routing
js/core.js	shared state, helpers, toasts
js/auth.js / js/firebase.js	sign-in / Firebase setup
js/data.js	every Firestore read/write + R2 helpers
js/shell.js	header + tab navigation
js/views/*.js	one file per page (dashboard, team, log, training, library, matches, leaderboard, mindroom, login)
js/court3d.js	the Pro Court engine (3D cameras, figures, timeline)
js/courtlab.js , js/sidecourt.js , js/solocourt.js	legacy court engines (old saved drills)
js/r2upload.js	browser → R2 upload with progress/dedupe
js/report.js	PDF report
api/r2-upload-url.js	presigned upload URL (any signed-in member)
api/r2-play-url.js	presigned playback URL (signed-in)
api/r2-delete.js	delete stored files (coach only)
api/_lib/	R2 client + Firebase admin token verification
firestore.rules	the database rules (publish via Firebase console)

Firestore collections: [members](#) (roster; doc ID = Auth UID), [measurements](#), [goals](#), [training](#), [plans](#) (per player + [__schedule](#), [__giftjar](#)), [drills](#), [videos](#), [private/{uid}](#) (mood stars), [messages](#) (player→coach).

Vercel environment variables: [R2_ACCOUNT_ID](#), [R2_ACCESS_KEY_ID](#), [R2_SECRET_ACCESS_KEY](#), [R2_BUCKET](#), [COACH_EMAILS](#), [MAX_UPLOAD_MB](#), [FIREBASE_SERVICE_ACCOUNT](#) (full service-account JSON).

Deploying a change: push to [main](#) → Vercel builds automatically. Database-rule changes additionally require pasting [firestore.rules](#) into Firebase Console → Firestore → Rules → Publish.

7. Common admin tasks

- **Add a team member:** Firebase Console → Authentication → Add user (fake email + password) → copy their UID → Firestore → `members` → new doc with **Document ID = that UID** and fields `id` (number), `name`, `role` (`player/coach`), `photo` (optional `imgs/Name.jpeg`).
 - **Change the tests:** edit `TESTS` in `js/config.js`.
 - **Update tournament venues/dates:** in-app — Training → 🏆 → 📅 Edit schedule.
 - **Add the coach's WhatsApp later:** set `COACH_CONTACT.whatsapp` in `js/config.js` (full international number, digits only).
 - **Lost R2 keys:** create a new R2 API token in Cloudflare, update the two key variables in Vercel, redeploy.
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8. Roadmap

Agreed next ideas (in rough order):

1. **Drill DNA label** — every drill auto-computes its training load (metres run, lunges, jumps, smashes, court-zone split, intensity).
2. **Playable drills** — players tap where they'd hit before the feed arrives; scored decision training with a Tactical-IQ leaderboard.
3. **On-court mode** — big step-by-step display + audible feed metronome at the chosen feed rate; optional coach voice per step.
4. **Shuttle-cam & player's-eye** camera presets.
5. **One-tap drill remix** — mirrored / roles swapped / faster variants.
6. **Film-room ↔ drill links** — pin a drill to a video moment: mistake → corrective drill in one tap.
7. Ongoing: ever-finer shot realism in the Pro Court.